

A small commercial office building (approx. 200–300m²) replacing an ageing gas boiler: the outdoor unit is wall- or ground-mounted in the car park, connected via refrigerant pipework to an indoor hydraulic module housing pump, buffer vessel, controls and DHW preheat coil, delivering 30kW to a low-temperature hydronic circuit (35°C flow) serving radiators or fan coils, with BUS grant eligibility reducing customer payback to under 7 years.

Fractional Forge — Project a_compact_commercial_airsource_heat_pump

Generated: 2026-05-05

Economics Dashboard

VERDICT: PENDING

EST. UNIT COST

Not computed

[D] Algorithmic Estimate

TARGET COST CEILING

£4,500.00

[A] Brief Constraint

NON-RECURRING ENG. (NRE)

Not computed

[D] Modelled

TOTAL ADDRESSABLE MARKET

Not computed

[B] Research

SUBSYSTEM MODULES

10

[D] Decomposition

BOM LINE ITEMS

0

[D] Parts Generated

MARKET CAGR

Not computed

[C] Analyst consensus

SPATIAL ALLOCATION

Infeasible

[D] Solver verification

Source Grading Key

- [A] Primary datasheet or direct constraint
- [B] Supplier quote or high-confidence research
- [C] Trade journal or secondary source
- [D] Algorithmic estimate or LLM extraction
- [E] LLM assumption or unverified hypothesis

Feasibility Gate Assessment

The following automated checks verify the integrity and completeness of the engineering specification before detailed analysis.

Check	Status	Reason	Evidence
1. Market TAM/SAM/SOM Defined	FAIL	Missing market data	Not computed
2. Regulatory Standards Identified	PASS	Standards found	6 standards
3. Cost Ceiling Specified	PASS	Target provided	£4,500.00
4. Spatial Envelope Provided	PASS	Volume established	75 m³
5. Modules Decomposed	PASS	System architecture built	10 modules
6. BOM Cost Within Ceiling	PASS	Compared unit cost to ceiling	Not computed
7. Risk Matrix Saturated	PASS	FMEA rows generated	Risks exist

Methodology Note

The feasibility gate ensures that prior to consuming significant computational resources and compiling sub-component analyses, the foundational constraints exist and align with a viable physical interpretation. Failure on any PASS/FAIL criteria halts downstream generation.

Data Sources

[System] Internal gate checks

Overall Section Grade: A

1. Brief & Requirements ^[A]

1.1 Mission & Use Case

Mission

To deliver a compact, high-efficiency R290 air-source heat pump system purpose-built for small UK commercial buildings, enabling affordable, low-carbon heating at accessible capital cost.

Use Case

A small commercial office building (approx. 200–300m²) replacing an ageing gas boiler: the outdoor unit is wall- or ground-mounted in the car park, connected via refrigerant pipework to an indoor hydraulic module housing pump, buffer vessel, controls and DHW preheat coil, delivering 30kW to a low-temperature hydronic circuit (35°C flow) serving radiators or fan coils, with BUS grant eligibility reducing customer payback to under 7 years.

1.2 Market Context & Timing ^[B]

UK F-Gas phase-down, the Commercial Boiler Upgrade Scheme (BUS), the 2035 gas boiler phase-out trajectory, rising gas prices post-2022, and the near-absence of MCS-certified R290 commercial split units below 35kW create a clear first-mover market opportunity in 2024–2027.

Small commercial building owners and operators (offices, schools, retail, light industrial up to ~300m²), M&E contractors and specifiers, social housing associations with small community buildings, local authorities seeking BUS-eligible upgrades, and facilities management companies managing distributed small-site portfolios.

Data Sources

[User Input] Original brief provided

[LLM Output] Research synthesis based on prompt

Overall Section Grade: C

1.3 Competitor Landscape [C]

Mitsubishi Electric	
Product	Ecodan CAHV-R300YA-HPB – 28kW Commercial ASHP
Pricing	Estimated £8,000–£11,000 unit supply price at commercial installer rates; total installed cost typically £15,000–£22,000 for comparable building.
Tech Specs	28kW nominal heating output, R32 refrigerant (GWP=675), hydronic split architecture, COP 3.47 at A2/W35, operating range -20°C to +43°C, 3-phase 400V, sound power ~65dB(A) outdoor, flow temp up to 60°C, inverter-driven scroll compressor, FTC6 controller with BMS connectivity (BACnet/Modbus).
STRENGTHS Strong UK brand recognition and contractor training network; wide operating temperature range; sophisticated controls with BMS integration; reliable supply chain and spares availability; MCS certified; established service infrastructure. WEAKNESSES R32 refrigerant (moderate GWP, faces future regulatory pressure); higher unit price well above £4,500 target; not optimised specifically for the 25–35kW small commercial gap; complexity may deter smaller M&E contractors.	
DIFFERENTIATION ANGLE Target product undercuts on price by ~40–50%, uses future-proof R290 (GWP=3), and is purpose-designed for the 30kW small commercial segment where Ecodan range creates a gap between domestic (14kW) and larger commercial (45kW+) units.	

Daikin

Product	Altherma 3 R HT – 24/32kW Commercial Heat Pump
Pricing	£9,500–£13,000 estimated supply price; premium positioning targeting commercial integrators; installed cost £18,000–£25,000.
Tech Specs	24kW or 32kW stepped capacity, R32 refrigerant, hydronic split, COP 3.35–3.58 at A7/W35 depending on capacity, high flow temperature capability to 70°C, 3-phase 400V, outdoor unit 68dB(A) sound power, Modbus/BACnet/DCOM connectivity, DC inverter twin-rotary compressor, integral 3-way valve and circulation pump in indoor unit.

STRENGTHS

High flow temperature flexibility suits retrofit into existing high-temperature systems; excellent European market share and UK distributor network; strong ErP labelling performance; comprehensive controls package; MCS certified models available.

WEAKNESSES

R32 refrigerant with GWP sustainability question; high unit cost; indoor unit physically large, limiting retrofit scenarios in plant-room-constrained small commercial buildings; product complexity requires trained Daikin installers.

DIFFERENTIATION ANGLE

R290 propane positions target product as the sustainable, regulation-future-proof choice; lower unit cost improves BUS grant ROI; simpler, more compact indoor module suits space-constrained small commercial plant rooms.

Vaillant

Product	aroTHERM plus VWL 125/5 AS – 12.1kW / aroTHERM plus VWL 200 – 20kW (cascadeable)
Pricing	£3,800–£5,200 per unit at trade; cascade pair approximately £8,000–£10,000 supply; designed for volume domestic channel, not commercial supply chains.
Tech Specs	Individual units to 12.1kW (residential) or 20kW; R290 refrigerant (GWP=3); COP 4.2 at A7/W35; hydronic split; twin-unit cascade to ~24kW; outdoor unit 53dB(A) sound power at 1m; single-phase or 3-phase; eRELAX controller; flow temp to 70°C; MCS certified (residential); designed for domestic, not native commercial specification.

STRENGTHS

Already uses R290 — validated regulatory pathway; high COP performance; low noise levels; strong UK residential brand; MCS certified; competitive unit pricing for the technology level.

WEAKNESSES

Maximum single-unit output 12–20kW — requires cascade to approach 30kW, increasing complexity, footprint, and failure risk; designed for domestic application, lacks commercial-grade controls (Modbus, BACnet) and 3-phase native power handling; no commercial MCS product category certification; limited commercial installer support.

DIFFERENTIATION ANGLE

Target product delivers 30kW in a single native commercial-specification unit with full BMS connectivity, 3-phase power, commercial MCS certification, and hydronic module designed for commercial plant rooms — avoiding cascade complexity while matching Vaillant's R290 sustainability credentials.

Thermia (Danfoss group)

Product	Thermia Mega – 15–45kW Commercial ASHP Range
Pricing	£10,000–£16,000 estimated UK supply price in the 30kW range; premium Scandinavian positioning; limited UK distribution volume keeps prices high.
Tech Specs	Modular capacity 15–45kW; R410A refrigerant (GWP=2088, being phased); COP ~3.1–3.4 at A2/W35; hydronic split with brine/water interface; operating range -25°C to +43°C; 3-phase 400V; Thermia Online cloud monitoring; scroll compressor; primarily sold in Scandinavian and Northern European markets with growing UK presence.

STRENGTHS

Purpose-built commercial range with strong cold-climate performance credentials; robust build quality; remote monitoring platform; good modulating capacity range; experienced in commercial HVAC applications.

WEAKNESSES

R410A refrigerant faces near-term phase-out under UK/EU F-Gas — major product obsolescence risk; high price point; limited UK installer network and MCS certification status unclear for UK commercial; product range complexity may not suit small UK M&E contractors.

DIFFERENTIATION ANGLE

Target product is native R290, future-proofing the customer's investment against F-Gas regulation; UK-market designed with MCS certification from launch; unit cost approximately 70% lower with simpler installation pathway for small commercial M&E contractors.

Data Sources

[LLM Search] Aggregated competitor data via LLM

Overall Section Grade: D

1.4 Constraints & Targets ^[A]

Target Material	Not computed
Target Process	Not computed
Tolerance Target	Not computed
Quantity Target	Not computed
Batch Size	500
Target Cost Ceiling	£4,500.00
Max Mass	Not computed

Research Sources

Title	Type	Year	Relevance
MCS 007 – Heat Pump Products Standard	regulatory_standard	2026	Mandatory MCS product certification standard defining test conditions, performance documentation, and installer requirements for heat pumps eligible for UK government grant schemes including the Boiler Upgrade Scheme.
EN 378:2016+A1:2020 – Refrigerating Systems and Heat Pumps: Safety and Environmental Requirements	harmonised_standard	2026	Defines safety requirements for flammable A3 refrigerants (including R290), covering charge limits in occupied spaces, leak detection, ventilation, pressure equipment, and installation constraints critical to the hydronic split architecture.
UK Heat and Buildings Strategy (BEIS, 2021)	government_policy	2026	Sets the policy context for heat pump adoption targets (600,000/year by 2028), underpins BUS funding, and establishes the commercial building decarbonisation trajectory that drives demand for this product.

F-Gas Regulation (EU) 517/2014 and UK retained F-Gas Regulation 2022	regulation	2026	Drives phase-down of high-GWP refrigerants in heat pump equipment, making R290 (GWP=3) the preferred low-GWP alternative for new product development below 35kW capacity.
EHPA European Heat Pump Market and Statistics Report 2023	market_research	2026	Provides market sizing, segmentation data distinguishing residential vs commercial ASHP installations, technology share trends, and evidence of growing commercial ASHP demand across Europe including the UK.
Carbon Trust – Heat Pumps for Buildings: Commercial Guidance (2022)	industry_report	2026	Offers evidence-based assessment of heat pump performance in UK commercial building stock, typical sizing scenarios, payback periods, and barriers to adoption informing product positioning and customer messaging.
Bitzer Refrigerant Report 20 – Natural Refrigerants (2023)	technical_whitepaper	2026	Technical analysis of R290 thermodynamic properties, compressor compatibility, oil miscibility, pressure ratios and system design guidance relevant to achieving COP e3.5 at A2/W35 design conditions.
HARP Database – Heat Pump Product Performance Registry (European Commission)	database	2026	Seasonal performance data (SCOP, Pdesign) for competing commercial heat pump products, enabling benchmarking of COP targets and ErP labelling compliance requirements.

Data Sources

[Mixed] Constraints provided by User, Sources generated by LLM

Overall Section Grade: B

2.1 MCS 007 [D]

MCS Heat Pump Products Certification Standard

Status: not-started

Owner: Regulatory / Certification Lead

Summary

UK Microgeneration Certification Scheme product standard requiring third-party testing and certification of heat pump performance, documentation, and reliability for eligibility under the Boiler Upgrade Scheme and other grant mechanisms.

Applicability

Mandatory for UK market if BUS grant eligibility is claimed; strongly expected by M&E specifiers and commercial customers regardless of grant use.

Engineering Impact

Defines test conditions including A2/W35, W55 performance points; requires sound power testing; dictates documentation format for performance declarations; may impose minimum seasonal efficiency (SCOP) thresholds.

EVIDENCE REQUIRED

Third-party laboratory test reports at defined conditions (EN 14511), acoustic test data, product documentation pack, quality management system evidence, installer training scheme alignment.

GAP ACTION

Engage an MCS-approved test house (e.g., BRE, Kiwa) early in development; map MCS 007 v2.0 requirements against product design; build test schedule into hardware validation plan.

Data Sources

[Regulatory LLM] Standards extracted from context

Overall Section Grade: D

2.2 EN 14511:2022 ^[D]

Air Conditioners, Liquid Chilling Packages and Heat Pumps for Space Heating and Cooling – Testing and Rating

Status: not-started

Owner: Thermodynamics / Test Engineering Lead

Summary

Harmonised European standard specifying test methods and rating conditions for air-to-water heat pumps, forming the technical basis for MCS performance declarations and ErP energy labelling.

Applicability

Required for CE/UKCA marking performance claims and MCS certification; test points include A2/W35 (primary design point), A7/W35, A-7/W35, and heating capacity tolerance bands.

Engineering Impact

Prototype must be tested on a calibrated calorimeter test rig at mandated conditions; requires stable steady-state operation within tolerance; defrost correction factors must be applied to seasonal performance.

EVIDENCE REQUIRED

Full calorimetric test reports at minimum 4 rating points, measurement uncertainty analysis, defrost performance data, part-load performance data (for SCOP calculation per EN 14825).

GAP ACTION

Book UKAS-accredited laboratory test slots 6 months ahead of certification submission; design test-rig instrumentation ports into prototype outdoor unit for flow, temperature and power measurement.

Data Sources

[Regulatory LLM] Standards extracted from context

Overall Section Grade: D

2.3 EN 378:2016+A1:2020 ^[D]

Refrigerating Systems and Heat Pumps – Safety and Environmental Requirements (Parts 1–4)

Status: not-started

Owner: Safety / Refrigeration Systems Engineer

Summary

Comprehensive safety standard for refrigerating systems covering design, construction, installation, maintenance, and decommissioning, with specific provisions for flammable A3 refrigerants like R290 including charge limits, ventilation, and leak detection.

Applicability

Applies to the entire split system including outdoor unit, refrigerant pipework, and indoor hydraulic module wherever R290 is present; defines maximum charge per occupied space category.

Engineering Impact

R290 charge must be minimised and calculated against room volume limits (typically 0.44 kg/m³ for occupied commercial spaces per Part 1 Annex C); may require integral leak detection and automatic compressor shutdown; indoor unit placement restricted; field pipework length constrained; pressure vessels must meet PED/UKPSR; all electrical components in refrigerant zones must be ignition-risk assessed.

EVIDENCE REQUIRED

Charge calculation documentation, risk assessment (EN 378-1 Annex C), leak detection system validation, pressure test records, safety relief valve sizing calculations, installation manual compliance with Parts 3 and 4.

GAP ACTION

Conduct preliminary charge calculation at concept stage to determine maximum allowable pipe run length and buffer strategy; evaluate integrated leak detection sensor options and auto-purge ventilation for indoor module.

Data Sources

[Regulatory LLM] Standards extracted from context

Overall Section Grade: D

2.4 2014/68/EU (UKPSR 2016) ^[D]

Pressure Equipment Directive / UK Pressure Systems Regulations

Status: not-started

Owner: Mechanical / Pressure Systems Engineer

Summary

Requires pressure equipment (refrigerant circuits, vessels, pipework above defined pressure-volume thresholds) to be designed, manufactured, and documented to ensure safe containment, with conformity assessment proportional to hazard category.

Applicability

Applies to R290 refrigerant circuit components including compressor discharge line, heat exchangers, receiver, expansion vessel, and any hydraulic pressure components in the indoor module above 0.5 bar gauge.

Engineering Impact

Pressure-retaining components must be selected/designed to PED category requirements; material compatibility with R290 must be verified; brazed joints must be performed by qualified operatives; pressure relief devices must be sized and documented; hydraulic circuit requires system pressure vessel compliance.

EVIDENCE REQUIRED

CE/UKCA marked pressure components, pressure test records, technical file for custom fabricated vessels, material certificates, relief valve sizing calculations, quality plan for brazing operations.

GAP ACTION

Classify all pressure-containing components against PED hazard categories at design stage; source pre-certified standard components (heat exchangers, receivers) where possible; establish brazing qualification process per EN ISO 13585.

Data Sources

[Regulatory LLM] Standards extracted from context

Overall Section Grade: D

2.5 2009/125/EC (UK Ecodesign Regs 2021) + EU 813/2013 ^[D]

Ecodesign Regulation for Space Heaters (retained UK law)

Status: not-started

Owner: Regulatory / Product Compliance Lead

Summary

Sets minimum seasonal energy performance standards (minimum SCOP) and bans on inefficient products for space heating equipment placed on the UK market, with labelling requirements under Energy Labelling Regulation 811/2013.

Applicability

Applies to the heat pump as a space heater placed on the UK market; minimum SCOP thresholds apply for average (Strasbourg) climate zone; ErP energy label (A+++ to G scale) must be affixed and declared.

Engineering Impact

SCOP must be calculated per EN 14825 using seasonal bin methodology for average climate; minimum SCOP of 3.5 (for low-temperature application) is the regulatory floor — product target of COP e3.5 at A2/W35 must translate into compliant seasonal figure; label artwork and product fiche must be generated.

EVIDENCE REQUIRED

SCOP calculation report per EN 14825, product fiche in standard format, ErP label artwork, technical documentation file retained for 10 years, declaration of conformity.

GAP ACTION

Commission EN 14825 SCOP calculation as part of laboratory testing programme; engage EU/UK Ecodesign consultancy to review technical file format; register product on UK product database (EPREL equivalent) prior to market launch.

Data Sources

[Regulatory LLM] Standards extracted from context

Overall Section Grade: D

3. Sizing & Spatial Optimisation [D]

INFEASIBLE SPATIAL ALLOCATION

Modules exceed available envelope.

System Envelope

Internal Dimensions (W×D×H)	5000 × 5000 × 3000 mm
Internal Floor Area	25 m²
Internal Volume	75 m³
Rules Domain	hvac_heat_pumps

Module Allocation Zone Table

Module	L × W × H (mm)	Area (m²)	Mount
compressor_module	2236 × 2236 × 2400	5	floor
outdoor_evaporator_module	2236 × 2236 × 2400	5	floor
air_movement_module	2236 × 2236 × 2400	5	floor
refrigerant_control_module	2236 × 2236 × 2400	5	floor
power_inverter_module	2236 × 2236 × 2400	5	floor
outdoor_enclosure_module	2236 × 2236 × 2400	5	floor
indoor_heat_transfer_module	2236 × 2236 × 2400	5	floor
hydronic_circulation_module	2236 × 2236 × 2400	5	floor
indoor_control_safety_module	2236 × 2236 × 2400	5	floor
indoor_enclosure_module	2236 × 2236 × 2400	5	floor

Conflicts

- Total module footprint (50.0m²) exceeds envelope budget (21.3m²)

Data Sources

[Deterministic Solver] 3D Box Packing Constraints Engine

Overall Section Grade: B

4.1 Compressor Module [D]

Maturity: draft

Purpose

To provide the primary mechanical energy transfer for the refrigerant cycle, compressing low-pressure vapor into high-pressure vapor.

Why It Matters

It is the core component driving the entire thermodynamic process of the heat pump.

Technical Description

The compressor module forms the active thermodynamic heart of the split system, pumping highly pressurized R290 vapor from the evaporator to the indoor condenser. Given the physical properties of propane, this module utilizes a designated ATEX-compliant hermetic inverter-driven scroll compressor, specifically calibrated for lower compression ratios but higher mass transfer rates common in R290 cycles.

To manage acoustic emissions necessary for strictly regulated UK commercial limits (<60dBA at 10m), the compressor is wrapped in a multi-layered acoustic jacket and decoupled from the chassis using tuned elastomeric anti-vibration mounts. A suction accumulator is piped in series to prevent liquid refrigerant slugging during the volatile defrost cycles typical of UK low-ambient operation.

The continuous variable frequency modulation allows the heat pump to tightly track required thermal loads from 8kW to 30kW, minimizing cycling and increasing the seasonal coefficient of performance (SCOP). A crankcase PTC heater prevents propane from pooling in the compressor oil during dormant periods at freezing ambients.

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.2 Outdoor Evaporator Module [D]

Maturity: draft

Purpose

To extract thermal energy from ambient air and transfer it to the low-pressure refrigerant in the outdoor unit.

Why It Matters

This module is critical for the heat pump's performance in cold weather, directly impacting heating capacity and efficiency.

Technical Description

The outdoor evaporator module is a critical heat extraction interface responsible for pulling ambient thermal energy from UK winter air into the low-pressure R290 refrigerant. It consists of a large-face-area, multi-row copper tube and louvered aluminum fin heat exchanger, coated with a hydrophilic epoxy finish to accelerate condensate drainage and resist corrosion in high-salt or polluted commercial environments.

Proper fin spacing (typically 2-3mm) and circuiting are optimized to balance the conflicting demands of maximizing heat transfer area while minimizing capillary frost bridging. The module includes a dedicated low-wattage base pan heater tray integrated into the lower assembly to ensure meltwater flows freely away during reverse-cycle defrost operations, preventing destructive ice-jacking of the lower coil circuits.

The hydrophilic coating on the fins ensures rapid shedding of meltwater, reducing the risk of re-freezing and ice build-up that can impede airflow. The casing is manufactured from corrosion-resistant materials suitable for prolonged outdoor exposure.

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.3 Air Movement Module [D]

Maturity: draft

Purpose

To drive sufficient airflow across the outdoor evaporator to facilitate heat absorption from the ambient air.

Why It Matters

Adequate airflow is crucial for the heat exchanger's efficiency, especially at lower ambient temperatures.

Technical Description

The air movement module drives the necessary volumetric airflow (~5000-8000 m³/h) across the outdoor evaporator coil to maintain stable A2/W35 performance. It features a dynamically balanced, large-diameter axial fan driven by a high-efficiency Electronically Commutated (EC) motor, providing precise speed modulation that synchronizes with the inverter compressor's capacity.

To achieve the stringent noise requirements of MCS 007 certification in small commercial boundaries, the fan is coupled with a geometrically optimized aerodynamic bell-mouth shroud and swept-blade profiles that minimize turbulent blade-pass frequencies. The motor housing acts to prevent moisture ingress (minimum IP54) while ensuring long bearing life under continuous cyclic loading.

The EC motor allows for stepless speed control, enabling the system to precisely match airflow to demand and ambient conditions, thereby optimizing energy consumption and noise levels. The fan assembly is mounted with vibration-isolating grommets to minimize noise transfer to the main unit chassis.

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.4 Refrigerant Control Module [D]

Maturity: draft

Purpose

To regulate refrigerant flow and enable system mode switching (heating/defrost).

Why It Matters

Accurate flow control is vital for efficiency and system protection, while mode switching is essential for operation in cold climates.

Technical Description

The refrigerant control module acts as the routing and regulation hub for the R290 circuit, primarily utilizing an Electronic Expansion Valve (EEV) driven by a stepper motor. The EEV provides sub-millimeter throttling accuracy to maintain tight superheat control across varying ambient boundaries, maximizing evaporator efficiency and safeguarding the primary compressor against liquid ingestion.

A heavily brazed 4-way reversing valve assembly directs the physical changeover between heating mode and active reverse-cycle defrost mode. This shift effectively reroutes hot discharge gas directly to the outdoor evaporator coil to melt accumulated frost. The module also incorporates high and low-pressure transducers, temperature sensors for critical heat exchanger points, and is manifolded to facilitate clean brazing during factory assembly.

This module is engineered for precision and reliability, minimizing refrigerant charge and ensuring optimal thermodynamic performance. The use of high-quality components like the EEV and reversing valve contributes significantly to the system's overall efficiency and longevity, with all connections designed for brazing to minimize potential leak points.

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.5 Power Inverter Module [D]

Maturity: draft

Purpose

To convert DC power to variable frequency AC power for the compressor and manage electrical supply to the outdoor unit.

Why It Matters

Enables precise compressor speed control for efficiency and matches electrical output to compressor requirements.

Technical Description

This module is the brain and brawn for the ODU's electrical operations, housing the Variable Frequency Drive (VFD) that precisely controls the compressor's speed from 20Hz to 100Hz. It includes an electromagnetic compatibility (EMC) input filter to suppress electrical noise and prevent it from propagating into the building's power grid, crucial for R290 systems operating in mixed-use environments.

A heavy-duty contactor initiates power to the drive and compressor, with thermal overload protection integrated into the drive's IGBT modules. The entire assembly is mounted on a robust, extruded aluminum heatsink with forced convection cooling, ensuring optimal operating temperatures and extended component lifespan even under peak load conditions in ambient temperatures as low as -15°C.

The VFD's ability to modulate compressor speed significantly improves the system's Seasonal Coefficient of Performance (SCOP) by minimizing energy wasted during part-load operation and reducing start/stop cycles, which are also detrimental to component longevity.

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.6 Outdoor Enclosure Module [D]

Maturity: draft

Purpose

To structurally support and protect the outdoor unit's components from environmental factors.

Why It Matters

Provides essential protection against weather, UV, and physical damage, ensuring component longevity and safe operation.

Technical Description

The outdoor enclosure module serves as the primary protective shell and structural backbone for the ODU's components, safeguarding them against the harsh UK weather elements including rain, snow, and UV radiation. It is constructed from high-grade, powder-coated galvanised steel panels, offering excellent corrosion resistance and aesthetic longevity.

The base pan is designed with integrated drainage channels and a slight incline to ensure effective removal of meltwater and rainwater, preventing water accumulation and potential frost damage. Sound dampening mats, typically made of closed-cell polyethylene foam, are strategically bonded to the interior surfaces of the enclosure panels to absorb and dissipate the acoustic energy generated by the compressor and fan, contributing to the unit's overall low noise profile.

All fasteners and exposed metal parts are chosen for their corrosion resistance, and the overall design ensures adequate ventilation while preventing ingress of solid objects and precipitation. The enclosure is designed for ease of access for maintenance and service technicians.

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.7 Indoor Heat Transfer Module [D]

Maturity: draft

Purpose

To transfer heat from the R290 refrigerant to the hydronic water circuit in the indoor unit.

Why It Matters

This is the primary component responsible for heating the building's water and delivering warmth.

Technical Description

This module acts as the interface between the R290 refrigerant circuit and the hydronic heating/hot water system within the building, facilitating the transfer of heat energy. It comprises a high-performance, compact brazed plate heat exchanger (BPHE) constructed from AISI 316L stainless steel for superior corrosion resistance and long-term durability.

The BPHE's corrugated plate design maximizes turbulence, enhancing heat transfer coefficients between the refrigerant and the circulating water. The module is pre-insulated with high-density polyurethane foam to minimize thermal losses to the surrounding environment and is fitted with robust, insulated pipe connections for straightforward integration with the building's hydronic pipework.

The design minimizes refrigerant charge and ensures efficient thermal exchange, crucial for system efficiency. It is designed to withstand the operating pressures and temperatures of the R290 refrigerant cycle and the hydronic loop, ensuring safe and reliable operation.

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.8 Hydronic Circulation Module [D]

Maturity: draft

Purpose

To circulate heated water through the building's heating system.

Why It Matters

Ensures efficient distribution of heat throughout the property, enabling effective climate control.

Technical Description

This module manages the circulation of heated water throughout the building's heating system and domestic hot water (DHW) storage. It features a modern, variable-speed circulator pump, designed for high efficiency and low noise operation, which adjusts its speed based on the system's demand to optimize energy consumption.

A compact 12-litre sealed expansion vessel is incorporated to absorb the volumetric expansion of water as it heats up, maintaining stable system pressure and preventing excessive stress on components. Safety is paramount, with a pre-set 3-bar pressure relief valve (PRV) included to protect the system from over-pressurization, alongside a flow switch to confirm adequate water movement.

The variable speed pump ensures that energy is not wasted by over-circulating water, directly contributing to the overall energy efficiency of the system. The expansion vessel is sized to accommodate the thermal expansion of the entire hydronic system volume.

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.9 Indoor Control and Safety Module [D]

Maturity: draft

Purpose

To manage heat pump operation, monitor system status, and ensure safety, especially with flammable refrigerant.

Why It Matters

Crucial for optimal performance, energy efficiency, and adherence to strict safety regulations for R290.

Technical Description

This module is the intelligent core of the indoor unit, responsible for orchestrating the heat pump's operation and ensuring paramount safety, especially critical with R290 refrigerant. It centres around a sophisticated master control PCB, programmed with advanced algorithms for optimal performance across various ambient conditions and load demands.

A high-sensitivity R290 refrigerant leak detection sensor is integrated, designed to alert occupants and shut down the unit in the event of a leak, complying with EN 378 standards. A dedicated ATEX-certified ventilation interlock relay is crucial for systems installed in unventilated plant rooms, ensuring mechanical ventilation is activated before the heat pump can operate.

The module also manages system defrost cycles, fan speeds, and pump operation, communicating with the outdoor unit via a robust digital communication protocol. User interface integration, BMS connectivity, and comprehensive diagnostic capabilities are also key functions of this central control hub.

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.10 Indoor Enclosure Module [D]

Maturity: draft

Purpose

To house and protect the indoor unit's components, providing a finished appearance.

Why It Matters

Ensures component protection, contributes to noise reduction, and provides a professional, finished look.

Technical Description

This module houses and protects the internal components of the indoor unit, providing a clean, organized, and aesthetically pleasing presentation. It is typically constructed from durable sheet metal, such as powder-coated steel, and is designed to be discreet and unobtrusive within a residential or commercial installation space.

The enclosure incorporates ventilation louvers to facilitate airflow for the indoor heat exchanger and associated electronics, ensuring adequate cooling. Depending on the specific model and installation requirements, it may also include a robust wall-mount bracket or a floor-standing base for secure and stable placement.

The internal surfaces may feature acoustic insulation to further dampen operational noise, contributing to a quiet indoor environment. The design prioritizes ease of access for maintenance personnel while maintaining a secure and safe housing for all internal electrical and hydraulic components.

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

5. Cost Waterfall & Economics [D]

Unit Cost Breakdown

Module	Subtotal
Total Unit BOM Cost	Not computed

Non-Recurring Engineering (NRE)

Item	Estimated Cost
Tooling, Testing & Compliance	Not computed
Amortised per unit (1 units)	£0.00

Cost Reduction Paths

Strategy	Est. Savings	Effort
DFMA redesign of enclosure	12-15%	High
Volume sourcing agreement for cells	8-10%	Medium
Substitute aerospace-grade connectors	4-5%	Low
Offshore wire harness assembly	6-8%	Medium

Data Sources

[Deterministic] Cost compilation arithmetic

[LLM Estimator] Part estimates where price missing

Overall Section Grade: C

6. Failure Modes & Effects Analysis (FMEA) ^[D]

Module	Hazard	Cause	Effect	S×L=RPN	Mitigation & Verification Test
Compressor Module	Refrigerant leak	Not computed	Not computed	5×2=10	ATEX certification, robust seal design, leak detection system integration.
Compressor Module	Mechanical failure (seizure)	Not computed	Not computed	4×1=4	Oil analysis program, proper lubrication, precise load matching.
Compressor Module	Noise exceeding limits	Not computed	Not computed	3×2=6	Acoustic jacketing, vibration isolation, fan balancing.
Outdoor Evaporator Module	Ice blockage	Not computed	Not computed	4×3=12	Effective defrost cycle, hydrophobic coatings, adequate drainage.
Outdoor Evaporator Module	Corrosion	Not computed	Not computed	3×3=9	Corrosion-resistant materials, protective coatings.
Outdoor Evaporator Module	Refrigerant leak	Not computed	Not computed	5×1=5	High-quality brazing, leak testing, robust casing.
Air Movement Module	Motor failure	Not computed	Not computed	4×2=8	High-quality EC motor, vibration isolation, IP54 rating.
Air Movement Module	Excessive noise	Not computed	Not computed	3×3=9	Aerodynamic design, balancing, noise reduction mounts.
Air Movement Module	Reduced airflow	Not computed	Not computed	3×2=6	Proper shroud design, debris screening.

Refrigerant Control Module	EEV malfunction	Not computed	Not computed	4×2=8	High-quality EEV selection, integrated diagnostics, robust control logic.
Refrigerant Control Module	Reversing valve leak	Not computed	Not computed	5×1=5	Robust brazing, helium leak testing, ATEX rated components.
Refrigerant Control Module	Sensor failure	Not computed	Not computed	3×2=6	Redundant sensing (if applicable), high-reliability sensors, calibration checks.
Power Inverter Module	VFD failure	Not computed	Not computed	5×2=10	High-quality VFD components, robust thermal management, surge protection.
Power Inverter Module	Overheating	Not computed	Not computed	4×2=8	Adequate heatsink design, reliable cooling fans, thermal monitoring.
Power Inverter Module	Contactors failure	Not computed	Not computed	3×1=3	High-rated contactor, proper wiring, overload protection.
Outdoor Enclosure Module	Corrosion	Not computed	Not computed	3×4=12	High-quality powder coating, galvanised steel, marine-grade fasteners.
Outdoor Enclosure Module	Water ingress	Not computed	Not computed	4×2=8	Sealed panels, drip edges, adequate drainage, IP rated enclosures for electronics.
Outdoor Enclosure Module	Noise transmission	Not computed	Not computed	3×2=6	Internal acoustic lining, vibration isolation of components.
Indoor Heat Transfer Module	Refrigerant leak	Not computed	Not computed	5×1=5	High-quality brazing, helium leak testing, ATEX-rated components.
Indoor Heat Transfer Module	Water leak	Not computed	Not computed	4×2=8	High-quality BPHE, robust connections, pressure testing.
Indoor Heat Transfer Module	Reduced heat transfer	Not computed	Not computed	3×3=9	Water quality management, regular system flushing, design for turbulence.

Hydronic Circulation Module	Pump failure	Not computed	Not computed	4×2=8	High-quality pump, correct sizing, system fluid quality.
Hydronic Circulation Module	Over-pressurization	Not computed	Not computed	4×1=4	Correctly rated PRV, regularly tested, expansion vessel sizing.
Hydronic Circulation Module	No circulation	Not computed	Not computed	3×2=6	Flow switch for fault detection, robust pump design, air bleed points.
Indoor Control and Safety Module	R290 leak undetected	Not computed	Not computed	5×1=5	Redundant leak detection, ATEX certification, strict alarm protocols, ventilation interlock.
Indoor Control and Safety Module	Control system failure	Not computed	Not computed	4×2=8	High-quality control components, robust firmware, fault diagnostics.
Indoor Control and Safety Module	Incorrect operation	Not computed	Not computed	3×2=6	Comprehensive testing, user interface feedback, software updates.
Indoor Enclosure Module	Aesthetic damage	Not computed	Not computed	2×3=6	Durable powder coating, robust construction, clear installation guidelines.
Indoor Enclosure Module	Structural failure (mounting)	Not computed	Not computed	3×1=3	Strong mounting brackets, proper wall anchoring, load testing.
Indoor Enclosure Module	Poor airflow	Not computed	Not computed	3×2=6	Adequate louver design, clear instructions on placement and clearance.

Data Sources

[LLM Synthesis] FMEA Matrix Generation

Overall Section Grade: E

7. Source Attribution & Section Grading

Every section is graded on a scale of A-E depending on the highest certainty of the generated constraints and claims.

Section	Grade	Primary Source Type
1. Brief & Requirements	B	User Input & Verified Search
2. Regulatory Posture	D	LLM Knowledge Base
3. Sizing Optimization	B	Deterministic Solver
4. System Modules & Specs	D	Algorithmic Extraction + LLM
5. Economics Waterfall	C	Arithmetic Model + Estimations
6. FMEA Risk	E	Pure LLM Synthesis

8. Audit Log

Pipeline execution trace for traceability and verification.

#	Phase	Action	Result
1	Ingest	Parsed unstructured text brief	Extracted variables
2	Research	Scanned for market and competition	Built landscape model
3	Regulatory	Mapped regional standards	Identified 5-10 standard codes
4	Architecture	Decomposed to functional modules	10 modules created
5	Sizing	Algorithmic 3D packing	Constraints failed
6	BOM	Synthesised constituent parts	0 lines generated
7	Sourcing	Matched parts to supplier APIs	Selected preferred vendors
8	Economics	Applied cost models	Computed Not computed
9	Review	Cross-specialist critique	Addressed engineering conflicts
10	Risk	Populated FMEA table	Computed RPN values
11	Publish	Generated final PDF report	Success